

MARGHERITA MARTORANA, PhD

 LinkedIn  GitHub  Google Scholar

SUMMARY — Knowledge and AI scientist with a PhD in Computer Science and experience in knowledge graphs, ontologies, semantic technologies and AI. My work focuses on knowledge representation, data integration and the development of technical solutions for complex and heterogeneous data. My experience spans both research and applied projects involving Python, machine learning, LLMs and data engineering, often in collaboration with technical and non-technical stakeholders.

PROFESSIONAL EXPERIENCE

Knowledge and AI Expert

Feb 2026 – Present

Clinical Operations Support - University Medical Center (Utrecht, NL)

- Responsible for the design of knowledge architectures and semantic models for clinical operations, supporting digital products and operational workflows.
- Develop ontologies and knowledge representations to structure and connect complex clinical and organisational information.
- Contribute to AI- and knowledge-driven solutions by translating operational needs into technical models, data structures and system requirements.
- Collaboration with clinical, technical and organisational stakeholders.

PostDoc in AI

Apr 2025 – Sep 2025

Knowledge & AI Group - Vrije Universiteit (Amsterdam, NL)

- Developed neurosymbolic AI methods for personal service robotics, combining knowledge graphs, ontologies, multimodal models and robotic perception.
- Built architectures to connect outputs from multimodal and vision-language models with structured knowledge for reasoning and action.
- Designed semantic models linking robot capabilities, tasks, environments and AI components to support interoperability across heterogeneous systems.

Data Scientist Fellow

Jan 2020 – May 2020

Apolitical (London, UK)

- Built ETL pipelines and APIs integrating heterogeneous data sources for analytics and reporting.
- Developed dashboards and explainable analytical workflows to support data-driven decision-making.
- Deployed and maintained services using GCP and Kubernetes in a fast-paced startup environment.

EDUCATION

Ph.D. in Computer Science

Jan 2021 – Mar 2025

User Centric Data Science Group - Vrije Universiteit (Amsterdam, NL)

Thesis: Semantic interpretation of dataless tables: a metadata-driven approach for Findable, Accessible, Interoperable and Reusable restricted access data.

- Developed semantic and data-processing methods for restricted-access datasets, including ontologies and data translation pipelines.
- Applied LLMs and semantic technologies to metadata enrichment, semantic annotation, controlled-vocabulary mapping and data integration.
- Designed evaluation methods and consistency metrics to assess metadata quality and system outputs.
- Developed reusable tools and software supporting interoperable and reusable data infrastructures.

M.Sc. Bioinformatics with Systems Biology

Sep 2018 - Sep 2019

Birkbeck University (London, UK)

B.Sc. Biological Sciences

Sep 2015 - June 2018

Royal Veterinary College (London, UK)

SKILLS

Knowledge Systems Knowledge graphs, ontology engineering, RDF, SPARQL, JSON-LD, semantic web technologies, metadata modelling, vocabularies, semantic interoperability, FAIR data

AI & ML LLMs, NLP, embeddings, semantic search, RAG, multimodal AI, VLMs, neurosymbolic AI, prompt engineering, AI evaluation

Programming Python (pandas, NumPy, scikit-learn, TensorFlow/Keras, spaCy, rdflib), SQL, R

Data & Cloud ETL pipelines, APIs, data translation workflows, data quality assessment, BigQuery, GCP, Azure, Docker, Kubernetes, Git, GitHub, GitLab

Collaboration Agile/Scrum, stakeholder engagement, cross-functional teamwork

Languages Italian (native), English (C2, full professional proficiency), Dutch (A2)

- Projects**
- 🔗 **MetaLinker** – LLM/RAG-based table metadata annotation.
 - 🔗 **MetaUnionSearch** – Metadata-based union search for restricted-access data.
 - 🔗 **bridging-bots** – Neurosymbolic LLM-ontology framework for robotics.
 - 🔗 **LLMs-topic-classification** – LLM-based metadata topic classification.
 - 🔗 **pyDSV** – CSV-to-RDF/JSON-LD metadata conversion tool.
 - 🔗 **OntoBOT** – Ontology for interoperable robot task and capability modelling.

PUBLICATIONS

- 2025 **An Ontology for Unified Modeling of Tasks, Actions, Environments, and Capabilities in Personal Service Robotics**
- 2025 **Bridging Bots: from Perception to Action via Multimodal-LMs and Knowledge Graphs**
- 2025 **Metadata-driven Table Union Search: Leveraging Semantics for Restricted Access Data Integration**
- 2024 **Column Vocabulary Association (CVA): semantic interpretation of dataless tables**
- 2024 **Zero-Shot Topic Classification of Column Headers: Leveraging LLMs for Metadata Enrichment**
- 2023 **Advancing data sharing and reusability for restricted access data on the Web: introducing the DataSet-Variable Ontology**
- 2022 **Aligning restricted access data with FAIR: a systematic review**
- 2022 **Analysing the Evolution of Community-Driven (Sub-) Schemas within Wikidata.**

LEADERSHIP & ACTIVITIES

Supervision – Supervised MSc theses in Artificial Intelligence and mentored students in research design, implementation, evaluation and scientific writing.

Grant Activity – Secured three local grants as Budget Owner of the Forum for Young Scientists and contributed to national and institutional research proposals, including NWO applications.

Section Editor – Journal of Open Research Software · 2025 – Present
Support editorial review of research software submissions, with a focus on reproducibility, software quality, documentation and scientific relevance.

Board Member – Computer Science PhD Council, Vrije Universiteit · 2023 – 2024
Represented PhD candidates and worked with department leadership on initiatives supporting the doctoral community.

Board Member and Budget Owner – Forum for Young Scientists, Vrije Universiteit · 2021 – 2024
Coordinated workshops and outreach activities for the university's early-career research community and managed the forum's budget.

ADDITIONAL TRAINING

Netherlands Research School for Information and Knowledge Systems (SIKS)
Data Science: From Data Lakes to Knowledge Graphs — Machine Learning and Optimisation — Research Methods and Methodology — Human-Centered AI — Social AI

eScience Center
Machine learning in Python with scikit-learn

International Semantic Web Summer School

Aurora Masterclass on Public Outreach for Doctoral Candidates

AI and Health Winter School